

ALIKHAN MUSSIN

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Technologies & Skills

- **Languages:** Python, SQL, JavaScript
- **Web Development:** React, FastAPI, Rest APIs, HTML, CSS
- **Data Bases & ORM:** SQLAlchemy, SQLite, MySQL, Neo4j
- **AI & Data:** Hugging Face Transformers, Sentence Transformers, NLP, Semantic Research
- **Tools:** Git, GitHub, VS code, Power BI

Education

Czech University of Life Sciences, BS in Information Technology

Relevant coursework: Databases, Object-Oriented Programming, Algorithm Development, Software Development, Applied Mathematics for IT, Statistics, System Analysis

Projects

OpsGraph AI - AI-Powered Incident Intelligence Platform

Python, FastAPI, React, SQLAlchemy, Hugging Face Transformers, SQLite, | Personal Project

- Built a full-stack incident intelligence platform that automatically **classifies operational incidents by category and severity** using zero-shot NLP and domain-specific decision rules.
- Implemented **semantic similarity** search to identify related historical incidents and reuse previous human-validated classifications.
- Developed a **human-in-the-loop feedback mechanism** where analyst corrections are **persisted** and **prioritized** when classifying similar future incidents.
- Built incident analytics including **category distribution, severity filtering, search, sorting, pagination** and **historical incident analysis**.
- Designed **REST APIs** with **FastAPI** and **SQLAlchemy** and an interactive **React** interface for incident analysis and investigation.

TraceLite — Lightweight Error Monitoring Platform

Python, FastAPI, SQLAlchemy, SQLite, React, Pytest, GitHub Actions | Personal Project

- Developed a **Python SDK** that captures application exceptions, stack traces, environment details, and sends structured events to a **FastAPI ingestion API**.
- Implemented persistent error storage and **SHA-256 fingerprinting** to group repeated exceptions and track occurrence counts instead of storing duplicates.
- Built a **React** dashboard for viewing grouped issues, application/environment context, occurrence counts, and full stack traces.
- Added automated unit and **API integration** tests with **pytest** and configured GitHub Actions to run the backend test suite on Pull Requests and pushes to **main**
- Developed features through **Git feature branches, Pull Requests, diff review, and merge workflows**.

CogniLock — AI-Resistant Human Verification System (Project Developer)

Artificial Intelligence, Cybersecurity, Human Verification, CAPTCHA | Team Project

- Contributed to the design and development of an **AI-resistant security** solution addressing vulnerabilities in traditional **CAPTCHA** systems.
- Developed a human-verification concept combining **cognitive challenges** with **physical movements** to distinguish legitimate users from automated systems.
- Collaborated with the project team to explore alternative authentication methods aimed at strengthening website protection against AI-driven automated attacks.

- Participated in designing and refining the solution, focusing on **usability**, **security**, and **resistance** to automated CAPTCHA-solving techniques.